



PHANTOM 4.8 // DIAGRAM 10 — SECURITY, RBAC MATRIX & GOVERNANCE ARCHITECTURE

GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2
EXHIBIT

ZERO-TRUST PERIMETER DEFENSE & AUTHENTICATION LIFECYCLE

JWT (HS256) Access Tokens (15m)	Argon2id Password Hashing	SSRF Metadata IP Protection
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7-ROLE GRANULAR ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

SYSTEM_ADMIN	Full system config, camera onboarding, user provisioning, global audit logs
POLICE_OFFICER	Live video monitoring, watchlist management, incident escalation, PCR dispatch
INVESTIGATOR	Case dossier generation, historical ANPR search, certified court evidence export
ANALYST	Statistical dashboards, traffic density heatmaps, model accuracy reviews
VIEWER	Read-only live video grid and GIS map viewing (zero mutation or export rights)
AUDITOR	Immutable security audit trail inspection and compliance reporting
AI_WORKER	Machine-to-machine inference ingestion and real-time telemetry broadcast

SECTION 65B IMMUTABLE AUDIT TRAIL & SSRF PROTECTION

Cryptographic Audit Logging - Logs every API mutation, login, and export - Captures Officer ID, Client IP, Timestamp - Cryptographic request body hash verification <code>audit_hash = sha256(user + action + ts + req)</code> Immutable append-only DB table	SSRF & Infrastructure Protection - Blocks cloud metadata IP: 169.254.169.254 - Blocks localhost / 127.0.0.1 stream injection - Strictly validates RTSP & WHEP schemes - Enforces CORS origin whitelisting SlowAPI rate limits: 120 req/min per IP
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TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- 7-Role RBAC: SYSTEM_ADMIN, POLICE_OFFICER, INVESTIGATOR, ANALYST, VIEWER, AUDITOR, AI_WORKER with FastAPI dependencies.
- SSRF protection strictly blocks cloud metadata IP (169.254.169.254) and localhost loopback stream injections.
- Immutable audit log records every user mutation with timestamp, client IP, and cryptographic request hash.